# Chapter 8

# Other Algebras

A slightly more sophisticated way to view a **group G** is to think of it as consisting of a set $G$ together with a binary operation $* : G \times G \to G$, a unary operation $^{-1} : G \to G$ and a constant $e \in G$,

$$\mathbf{G} = \langle G; *, ^{-1}, e \rangle,$$

where $*$ is associative, $e$ is an identity, and $x^{-1}$ is an inverse of $x$ for each $x \in G$. It turns out that there are many different familiar algebraic systems that are similar to groups and can be studied in much the same way that we have studied groups. An **algebra**

$$\mathbf{A} = \langle A; \circ, *, +, \ldots, ^{-1}, \alpha, \neg, \ldots, e, 0, 1, \ldots \rangle$$

consists of a set $A$, a collection $\circ, *, +, \ldots$ of binary operations on $A$, a collection $^{-1}, \alpha, \neg, \ldots$ of unary operations on $A$, and a set $e, 0, 1, \ldots$ of distinguished constants from $A$.

Many central notions from group theory extend to all algebras. For example, an algebra

$$\mathbf{B} = \langle B; \circ, *, +, \ldots, ^{-1}, \alpha, \neg, \ldots, e, 0, 1, \ldots \rangle$$

is a **subalgebra** of $\mathbf{A}$ if it is closed under each of the operations of $\mathbf{A}$, that is,

- if $x, y \in B$ and $*$ is a binary operation of $\mathbf{A}$, then $x * y \in B$;
- if $x \in B$ and $'$ is a unary operation of $\mathbf{A}$, then $x' \in B$, and
- if $e$ is a constant of $\mathbf{A}$, then $e \in B$.

Notice that a subset of a group forms a subgroup in the usual sense if and only if it is a subalgebra in this sense.

Similarly, a function $h$ from an algebra $\mathbf{A}$ to an algebra $\mathbf{B}$ is a **homomorphism** if it preserves each of the operations, that is,

- $h(x * y) = h(x) * h(y)$ for each binary operation $*$ and all $x, y \in A$;
- $h(x') = h(x)'$ for each unary operation $'$ and all $x \in A$;
- $h(e_A) = e_B$ where $e_A$ is a constant of $\mathbf{A}$ and $e_B$ is the corresponding constant of $\mathbf{B}$.

Algebras $\mathbf{A}$ and $\mathbf{B}$ are **isomorphic** if there is a bijection $h : \mathbf{A} \to \mathbf{B}$ which is a homomorphism.

As we did with groups, we like to study classes of algebras with a fixed type of operations satisfying a fixed set of axioms. In this chapter we will introduce three such examples.

## RINGS

Our childhood experience with mathematics begins with the arithmetic of the integers and the rational numbers. These are two of many familiar examples of another kind of algebra. A **ring with identity** is an algebra $\mathbf{R} = \langle R; +, \cdot, -, 0, 1\rangle$ with two binary operations $+, \cdot$, a unary operation $-$ and two constants 0 and 1 such that

1. $\langle R; +, -, 0\rangle$ is a commutative group,
2. for all $x, y, z \in R$,
   - $x \cdot (y \cdot z) = (x \cdot y) \cdot z$,
   - $1 \cdot x = x = x \cdot 1$,
   - $x \cdot (y + z) = x \cdot y + x \cdot z$ and $(y + z) \cdot x = y \cdot x + z \cdot x$.

The third condition is expressed in words by saying that, "The operation $\cdot$ **distributes** over $+$." Note that since $\cdot$ may not be commutative, we must verify both distributive properties.

**Lemma 98.** *If* $\mathbf{R}$ *is a ring with identity and* $x \in R$*, then* $x \cdot 0 = 0 = 0 \cdot x$ *and* $-x = (-1) \cdot x$.

**Problem 99.** *Assume that* $\mathbf{R}^{+\cdot}$*, the real numbers with the usual operations, is a ring with identity. Show that each of the following is a ring with identity.* (*In each case you will need to tell what* 0 *and* 1 *are.*)

$\mathbf{Z}$*, the set of integers with the usual operations.*

$\mathbf{C}$*, the set of complex numbers with the usual operations,* $(a + bi) + (c + di) := (a + c) + (b + d)i$ *and* $(a + bi) \cdot (c + di) := (ac - bd) + (bc + ad)i$.

$\mathbf{M}$*, the set of 2-by-2 matrices over* $\mathbf{Z}$ *with the usual matrix addition and multiplication.*

$\mathbf{P_S} := \langle \mathbf{P}_S; \oplus, \cap, -, \varnothing\rangle$ *where* $S$ *is a set,* $\mathbf{P}_S$ *is the collection of subsets of* $S$*, and* $-X := X$ *for* $X \in P_S$.

$\mathbf{F}$*, the set of real valued functions defined on the set of real numbers, with the usual addition and multiplication,* $(f + g)(x) := f(x) + g(x)$ *and* $(f \cdot g)(x) := f(x)g(x)$.

**Problem 100.** *Show that* $\mathbf{Z}$ *is a subring of* $\mathbf{R}^{+\cdot}$ *which is a subring of* $\mathbf{C}$.

**Problem 101.** *Let* $a$ *be a real number, and define* $h_a : \mathbf{F} \to \mathbf{R}^{+\cdot}$ *by* $h_a(f) := f(a)$*. Show that* $h_a$ *is a ring homomorphism.*

**Problem 102.** *Let* $T$ *be a subset of a set* $S$ *and define* $h : \mathbf{P_S} \to \mathbf{P_T}$ *by* $h(X) := X \cap T$*. Show that* $h$ *is a ring homomorphism. Show that the multiplication operation* $\cap$ *is well defined on the set of cosets* $\mathbf{P_S}/\mathbf{K}_h$*, and that* $\mathbf{P_S}/\mathbf{K}_h$ *is itself a ring with identity that is isomorphic to* $\mathbf{P_T}$.

## LINEAR SPACES

In a college Linear Algebra course we learn about systems called "linear spaces". A **linear space** is an algebra $\mathbf{L} = \langle L; +, -, \mathbf{0}, a\rangle_{a \in R}$ with a binary operation $+$, a unary operation $-$, and a unary operation $a$ for each real number $a \in R$ such that

1. $\langle L;+,-,\mathbf{0}\rangle$ is a commutative group,
2. for all $X,Y \in L$ and $a,b \in R$,
   - $a(X+Y) = aX + aY$,
   - $(a+b)X = aX + bX$,
   - $a(bX) = (ab)X$ and
   - $aX = \mathbf{0}$ if and only if $a = 0$ or $X = \mathbf{0}$.

The study of linear spaces is called **linear algebra**. There are many familiar linear spaces: the Euclidean plane $\mathbf{R}^2$, Euclidean 3-space $\mathbf{R}^3$, Euclidean $n$-space $\mathbf{R}^n$, the space of real valued functions on any fixed domain, the space of solutions to a homogeneous differential equation, the space of $n$-by-$m$ matrices, etc.

A subalgebra of a linear space $\mathbf{L}$ is called a **subspace**. The smallest subalgebra of $\mathbf{L}$ containing a set $S \subseteq L$ is called the **span** of $S$, written $\mathrm{Span}(S)$. A generating set for $\mathbf{L}$ is called a **spanning set**. A homomorphism from one linear space to another is called a **linear transformation**.

**Theorem 103.** *Let* $\mathbf{L}$ *be a linear space, let* $\mathbf{M}$ *be any subspace of* $\mathbf{L}$*, and let* $\mathbf{L}/\mathbf{M}$ *be the set of right cosets of* $\mathbf{M}$ (*which forms a group under* $+$ *since* $\mathbf{M}$ *is a normal subgroup*)*. Then scalar multiplication is always well defined on* $\mathbf{L}/\mathbf{M}$ *as*

$$a(\mathbf{M}+X) := \mathbf{M}+aX,$$

*and* $\langle \mathbf{L}/\mathbf{M};+,-,0,a\rangle_{a\in R}$ *is itself a linear space* (*called the* **quotient space**).

**BOOLEAN ALGEBRA**

Modern algebra became a viable subject beyond group theory in the 1850's when George Boole showed how the laws of logical inference can be codified into an algebraic system in which the elements are statements. What we have come to call a **Boolean algebra** is an algebra $\mathbf{B} = \langle B;\vee,\wedge,',0,1\rangle$ with two binary operations **join** $\vee$ and **meet** $\wedge$, a unary operation of **complementation** $'$, and constants 0 and 1 such that, for all $x,y \in B$,

1. $\vee$ and $\wedge$ are both commutative and associative, and each distributes over the other,
2. $x \vee x = x$ and $x \wedge x = x$,
3. $x \vee (y \wedge x) = x$ and $x \wedge (y \vee x) = x$,
4. $x \vee 1 = 1$ and $x \wedge 0 = 0$,
5. $x \vee x' = 1$ and $x \wedge x' = 0$.

There are two important and familiar examples of Boolean algebras.

Let $S$ be a set and let $\mathbf{P}(S)$ denote the collection of all subsets of $S$. Then $\cup$ and $\cap$ are binary operations on $\mathbf{P}(S)$, complementation $\sim$ is a unary operation on $\mathbf{P}(S)$, and $\varnothing$ and $S$ are special elements of $\mathbf{P}(S)$. The proof of the following theorem has many parts, all of which are normally done in a foundations course in mathematics.

**Theorem 104.** *If* $S$ *is a set, then* $\mathbf{P}(\mathbf{S}) := \langle \mathbf{P}(S);\cup,\cap,\sim,\varnothing,S\rangle$ *is a Boolean algebra.*

Formulating propositional logic as a Boolean algebra requires an extra step that we draw from our experience with quotient groups. Consider any fixed set of propositional variables $p$, $q$, $r$,..., each representing a fixed statement. If we use $\vee$ for "or", $\wedge$ for "and", $'$ for "not", 0 for "False" and 1 for "True", we can combine propositional variables to for compound sentences such as

$$((p \vee 1) \wedge r)' \vee (q' \wedge (p \vee r)).$$

If $C$ denotes the set of all compound sentences, then $\vee$ and $\wedge$ are binary operations on $C$ while $'$ is a unary operation and 0 and 1 are constants in $C$. This gives us an algebra

$$\mathbf{C} := \langle C; \vee, \wedge, ', 0, 1 \rangle.$$

Is **C** a Boolean algebra? For example, is $\wedge$ commutative? We have $(q \vee r), p \in C$ so, for commutativity, we would need to know that

$$(q \vee r) \wedge p = p \wedge (q \vee r).$$

Is this the case? While these two compound sentences seem to make the same assertion ("$p$ is true and either $q$ or $r$ is true."), they are not really the same member of $C$. So $\wedge$ is not commutative. But the two compound sentences are equivalent in a natural sense: they have the same true tables.

These observations lead us to the following construction. For $x, y \in C$, we say that $x \equiv y$ if $x$ and $y$ have the same truth tables. We check that $\equiv$ is an equivalence relation and define

$$B = C/\equiv = \{[x] \mid x \in C\}$$

to be the set of equivalence classes of $C$.

**Theorem 105.** *The representative operations $\vee$, $\wedge$ and $'$ are well defined on $B$ by $[x] \vee [y] := [x \vee y]$, $[x] \wedge [y] := [x \wedge y]$ and $[x]' := [x']$, and the quotient algebra*

$$\mathbf{B} := \langle B; \vee, \wedge, ', [0], [1] \rangle$$

*is a Boolean algebra.*

These operations on the equivalence classes are well defined in exactly the same sense that a group operation is well defined on the cosets of a normal subgroup. The resulting Boolean algebra **B** is called the **Lindenbaum-Tarski algebra** for propositional logic. It allows us to understand mathematical logic through algebra.